[10, 46]

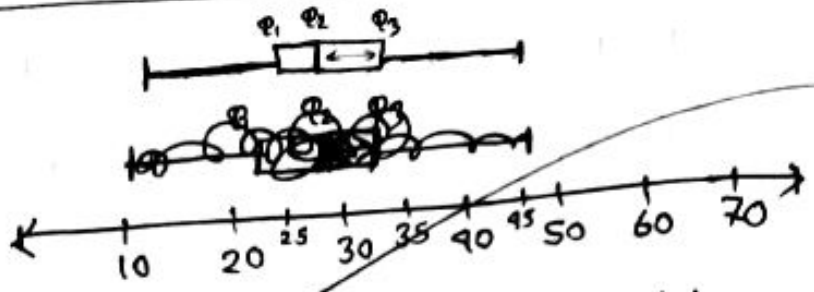


Fig: Box-and-whisker plot

ii) $Q_1 - Q_2 < Q_3 - Q_2$

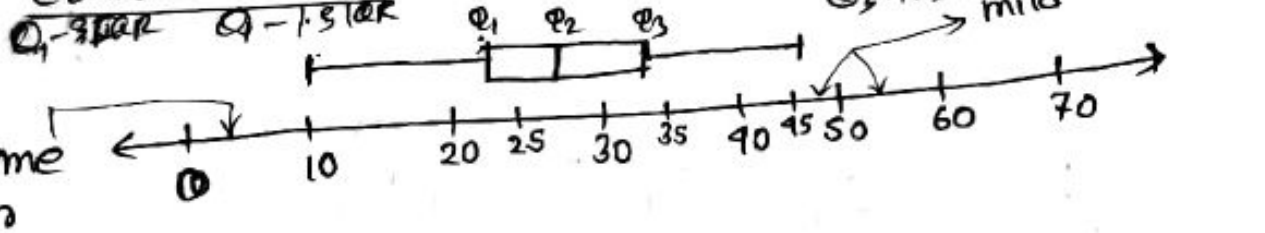
$4 < 5$



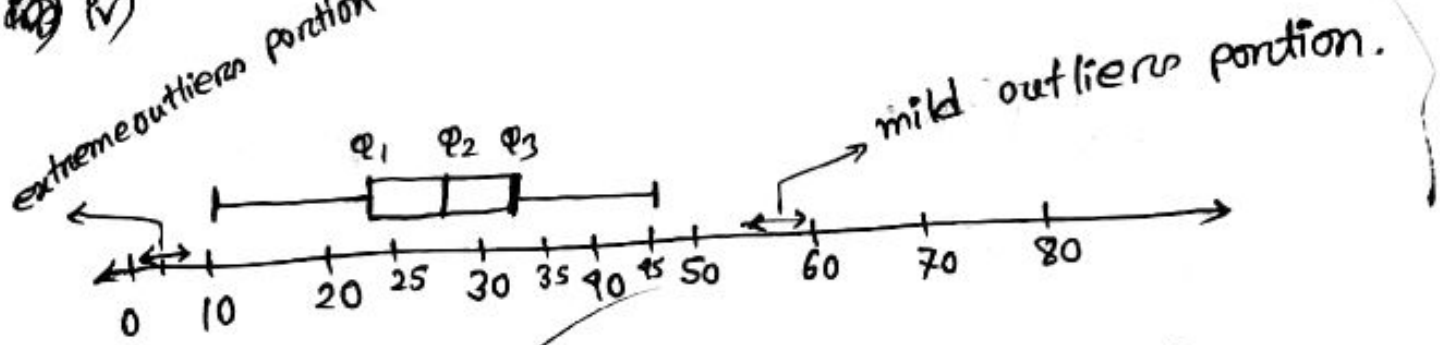
So these data are skewed in positive direction.

iii) The mild outliers are 53, 53, 60

The extreme outliers are 1, 1, 3, 4, 5, 8



iv)



(iv) Portion = ?

Ans to Q. No-2

1, 1, 3, 4, 5, 8, 17, 23, 24, 24, 25, 25, 25, 26, 26, 27, 28, 28, 28, 29, 29, 29, 29, 30, 35, 37, 40, 45, 45, 53, 53, 60

i) n is even $\quad n = 32$

$$Q_1 = \frac{1}{2}\left[x_{\left(\frac{32}{4}\right)} + x_{\left(\frac{32}{4}+1\right)}\right]$$

$$= \frac{1}{2}\left[x_{(8)} + x_{(9)}\right]$$

$$= \frac{1}{2}[23 + 24]$$

$$\therefore Q_1 = 23.5$$

$$\therefore Q_3 = \frac{1}{2}\left[x_{\left(\frac{3\times32}{4}\right)} + x_{\left(\frac{3\times32}{4}+1\right)}\right]$$

$$= \frac{1}{2}\left[x_{(24)} + x_{(25)}\right]$$

$$= \frac{1}{2}[30 + 35]$$

$$\therefore Q_3 = 32.5$$

$$\therefore Q_2 = \frac{1}{2}\left[x_{\left(\frac{2\times32}{4}\right)} + x_{\left(\frac{2\times32}{4}+1\right)}\right]$$

$$= \frac{1}{2}\left[x_{(16)} + x_{(17)}\right]$$

$$= \frac{1}{2}[27 + 28]$$

$$\therefore Q_2 = 27.5$$

$$\therefore IQR = Q_3 - Q_1$$

$$= 32.5 - 23.5$$

$$\therefore IQR = 9$$

$$\therefore \left[Q_1 - 1.5\,IQR,\ Q_3 + 1.5\,IQR\right]$$

$$\left[23.5 - 1.5\times9,\ 32.5 + 1.5\times9\right]$$

ii)

$$\beta_2 = \frac{\mu_4}{\mu_2^2} \quad , \quad \gamma_2 = \beta_2 - 3$$

$$\mu_2 = \frac{1}{N} \sum_{i=1}^{k} f_i (x_i - \bar{x})^2$$

$$= \frac{1}{659} \sum_{i=1}^{5} f_i (x_i - \bar{x})^2 = \frac{1}{659} \times 1315.85$$

$$\mu_2 = 2.012$$

$$\mu_4 = \frac{1}{N} \sum_{i=1}^{k} f_i (x_i - \bar{x})^4$$

$$= \frac{1}{659} \times 8545.75$$

$$\mu_4 = 13.07$$

$$\therefore \beta_2 = \frac{13.07}{(2.012)^2}$$

$$\therefore \beta_2 = 3.23$$

$$\bar{x} = \frac{1}{659} \sum_{i=1}^{k} f_i x_i$$

$$= \frac{1}{659} \times 1689.735$$

$$\therefore \bar{x} = 2.584$$

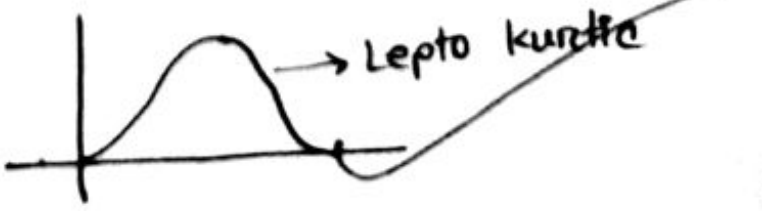
$$\gamma_2 = 3.23 - 3$$

$$\therefore \gamma_2 = 0.23$$

iii)

$\therefore \beta_2 > 3, \quad \gamma_2 > 0$

So the graph wave will be Lepto kurtic.



Ans to Q No-1

class interval	freq f_i	mid x_i	cu freq	$f_i x_i$	$(x_i - \bar{x})$
0.00-0.89	63	0.445	63	28.035	-2.139
0.89-1.78	128	1.335	191	170.88	-1.249
1.78-2.67	209	2.225	400	465.025	-0.359
2.67-3.56	106	3.115	506	330.19	0.531
3.56-4.45	76	4.005	582	304.38	1.421
4.45-5.35	41	4.9	623	200.9	2.316
5.35-6.24	19	5.795	642	110.105	3.211
6.24-7.13	12	6.685	654	80.22	4.101

$\Sigma f_i x_i = 1689.735$

1) Mode, $M_\theta = L + \dfrac{h(f_m - f_p)}{(f_m - f_p) + (f_m - f_a)}$

$= 1.78 + \dfrac{0.89(209 - 128)}{(209 - 128) + (209 - 106)}$

$= 1.78 + \dfrac{72.09}{184}$

$\therefore$ Mode $= 2.172$

Mode class
1.78 - 2.67

$L = 1.78$

$h = 0.89$

$f_m = 209$

$f_p = 128$

$f_a = 106$

$\therefore r_{xy} = \frac{17024.4}{\sqrt{20586.4 \times 14435.73}}$

$\therefore r_{xy} = 0.99$

coefficient of determination,

$R^2 = 0.975$

0.99

As $|r| =$, So there exists a very positive directional strongly correlation between the variables. And it is a linear correlation.

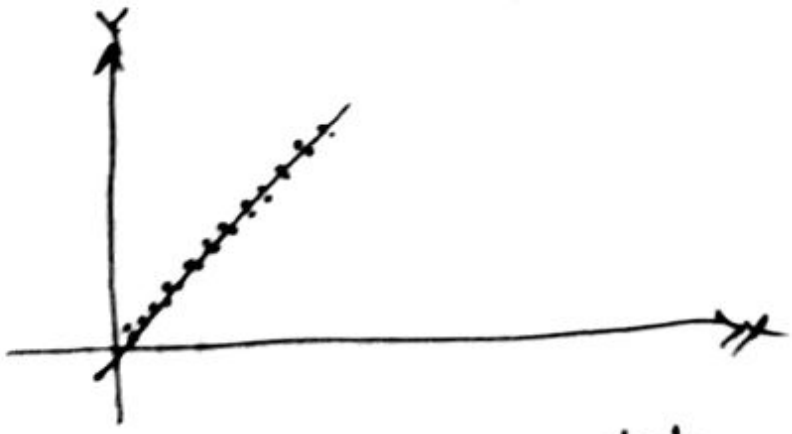


fig: Scatter plot

b)

When Rainfall volume = 50

$\therefore$ Runoff volume = ~~16.023~~ $-1.29 + ~~0.6099~~\ 0.83 \times 50$

$= ~~46.518~~\ 40.21\ (m^3)$

A

c) Correlation coefficient, $r_{xy} = \dfrac{SP(XY)}{\sqrt{SS(X)\ SS(Y)}}$

$\therefore SP(XY) = \sum_{i=1}^{n} x_i Y_i - n\bar{X}\bar{Y}$

$= 51232 - 15 \times \dfrac{798}{15} \times \dfrac{~~727~~\ 643}{15}$

$\therefore SP(XY) = ~~12555.6~~\ 17029.4$

$SS(X) = \sum_{i=1}^{n} x_i^2 - n\bar{X}^2 = 63040 - 15 \times \left(\dfrac{798}{15}\right)^2$

$\therefore SS(X) = 20586.9$

$SS(Y) = \sum_{i=1}^{n} Y_i^2 - n\bar{Y}^2 = 41999 - 15 \times \left(\dfrac{~~727~~\ 643}{15}\right)^2$

$\therefore SS(Y) = ~~6763.73~~\ 14435.73$

$$\beta_1 = \frac{\sum_{i=1}^{n} x_i y_i - \frac{\sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{n}}{\sum_{i=1}^{n} x_i^2 - \frac{(\sum_{i=1}^{n} x_i)^2}{n}}$$

$$= \frac{51232 - \frac{798 \times 643}{15}}{63040 - \frac{(798)^2}{15}}$$

$$= \frac{12555.6}{20586.4} \quad \frac{17029.9}{}$$

$\therefore \beta_1 = 0.6099 \quad 0.83$

$\bar{Y} = \beta_0 + \beta_1 \bar{x}$

$\therefore \beta_0 = \bar{Y} - \beta_1 \bar{x}$

$$= \frac{643}{15} - 0.6099 \times \frac{798}{15} \qquad 0.83$$

$= 48.47 - 32.447$

$\therefore \beta_0 = 16.023 \quad -1.29$

$\therefore$ Regression line, $\hat{Y} = 16.023 + 0.6099X$ $\quad -1.29 \quad 0.83$

$\therefore$ Runoff volume $= 16.023 + 0.6099$ Rainfall volume.

$-1.29 + 0.83$

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OBTAINED MARK 28

TUTORIAL ANSWER SCRIPT

Student's ID No: 359 **Name:** Trisha Sarkar

Course Code: CSE-251 **Course Title:** Introduction to Probability & statistics

Tutorial Examination No: 01 **Date:** 24-11-2022 **Signature of Course Teacher:** _____

Ans to Q-No-3

a)

X	Y	XY	x^2	Y^2
5	4	20	25	16
12	10	120	144	100
14	13	182	196	169
17	15	255	289	225
23	15	345	529	225
30	25	750	900	625
40	27	1080	1600	729
47	46	2162	2209	2116
55	38	2090	3025	1444
67	46	3082	4489	2116
72	53	3816	5184	2809
81	70	5670	6561	4900
96	82	7872	9216	6724
112	99	11088	12544	9801
127	100	12700	16129	10000
$\Sigma x = 798$	$\Sigma y = 693$	$\Sigma xy = 51232$	$\Sigma x^2 = 63040$	$\Sigma y^2 = 41999$